

Definition of e-commerce

E-commerce is an abbreviation used for electronic commerce. It is the process through which the buying, selling, dealing, ordering and paying for the goods and services are done over the internet is known as e-commerce. In this type of online commercial transaction, the seller can communicate with the buyer without having a face to face interaction.

Definition of e-business

Electronic Business, shortly known as E-business, is the online presence of business. It can also be defined as the business which is done with the help of internet or electronic data interchange i.e. is known as E-business. E-commerce is one of the important components of E-business, but it is not an essential part.

E-business is not confined to buying and selling of goods only, but it includes other activities that also form part of business like providing services to the customers, communicating with employees, client or business partners can contact the company in case if they want to have a word with the company, or they have any issue regarding the services, etc. All the basic business operations are done using electronic media.

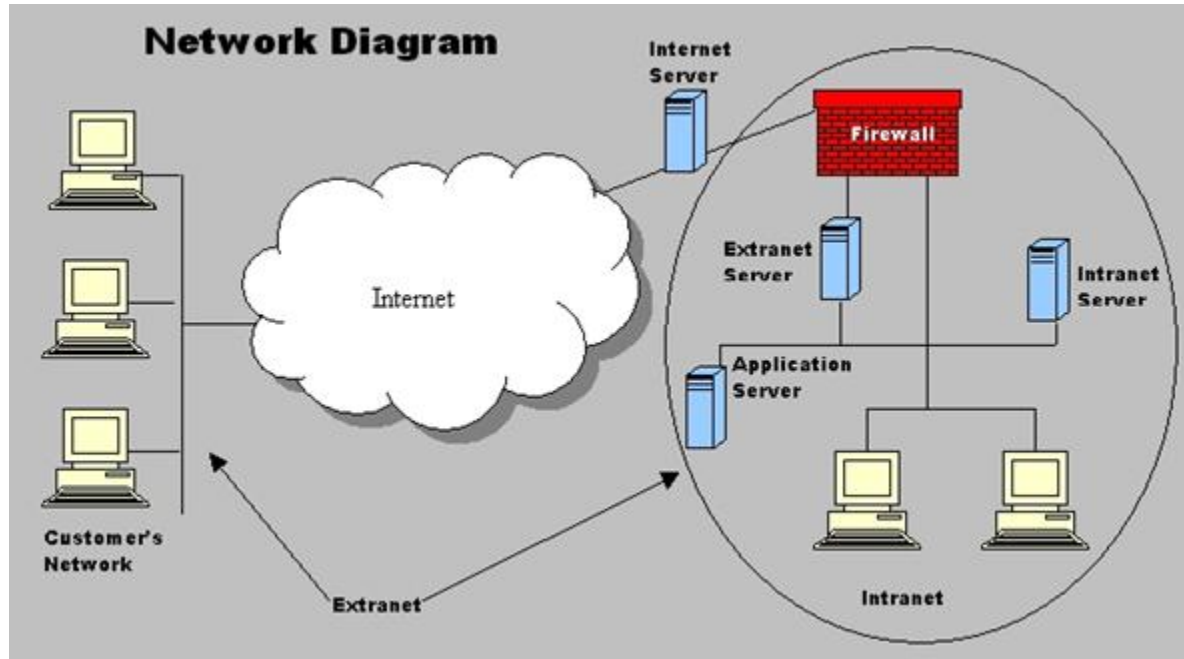
Key Differences Between e-commerce and e-business

The points presented below are substantial so far as the difference between e-commerce and e-business is concerned:

- 1.E-Commerce is buying and selling of goods electronically whereas E-Business not only includes buying and selling but it also includes CRM(customer resource management), ERM(enterprise resource management), etc.
2. E-commerce has an extroverted approach that covers customers, suppliers, distributors, etc. On the other hand, E-business has an ambivert approach that covers internal as well as external processes.
3. E-commerce uses the internet to connect with the rest of the world. In contrast to E-business, the internet, intranet and extranet are used for connecting with the parties.
- 4.E-Commerce involves commercial transactions whereas E-Business involves commercial as well as non-commercial transactions.
5. E-commerce requires a website that can represent the business. Conversely, E-business requires a website, Customer Relationship Management and Enterprise Resource Planning for running the business over the internet.
- 6.E-Commerce is part of e-business whereas E-Business includes E-Commerce.

- **Intranet** is shared content accessed by members within a single organization.
- **Extranet** is shared content accessed by groups through cross-enterprise boundaries.
- **Internet** is global communication accessed through the Web.

For better comprehension, take a look at this drawing:

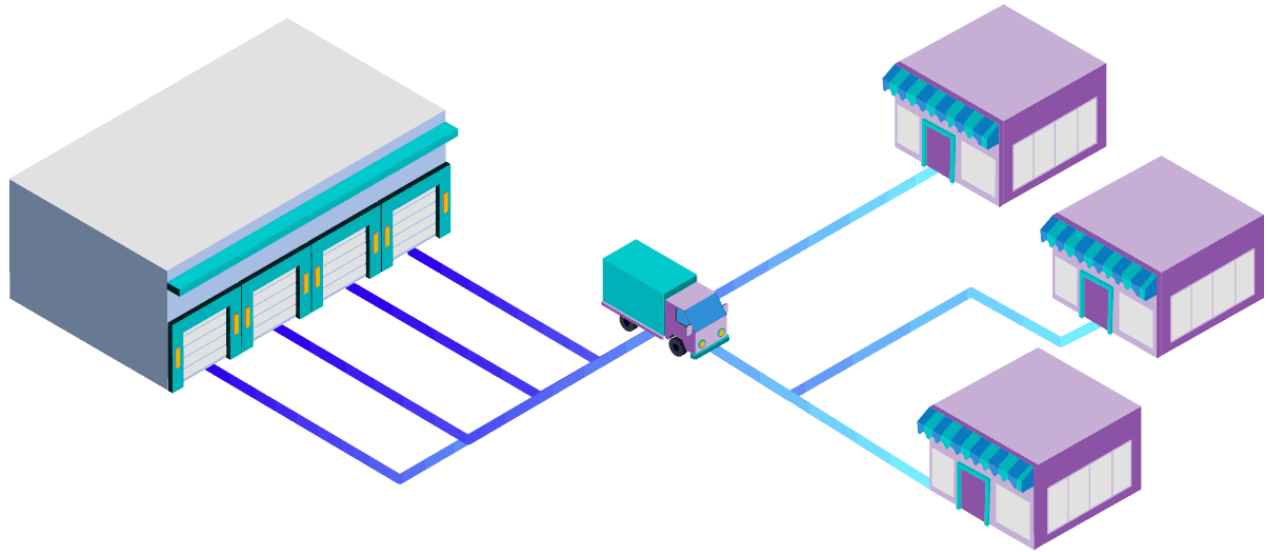


Summary:

The Internet, extranets, and intranets all rely on the same TCP/IP technologies. However, they are different in terms of the levels of access they allow to various users inside and outside the organization and the size of the network. An intranet allows for restricted access to only members of an organization; an extranet expands that access by allowing non-members such as suppliers and customers to use company resources. The difference between the Internet and extranets is that while the extranet allows limited access to non-members of an organization, the Internet generally allows everyone to access all network resources.

B2B: Business To Business Ecommerce

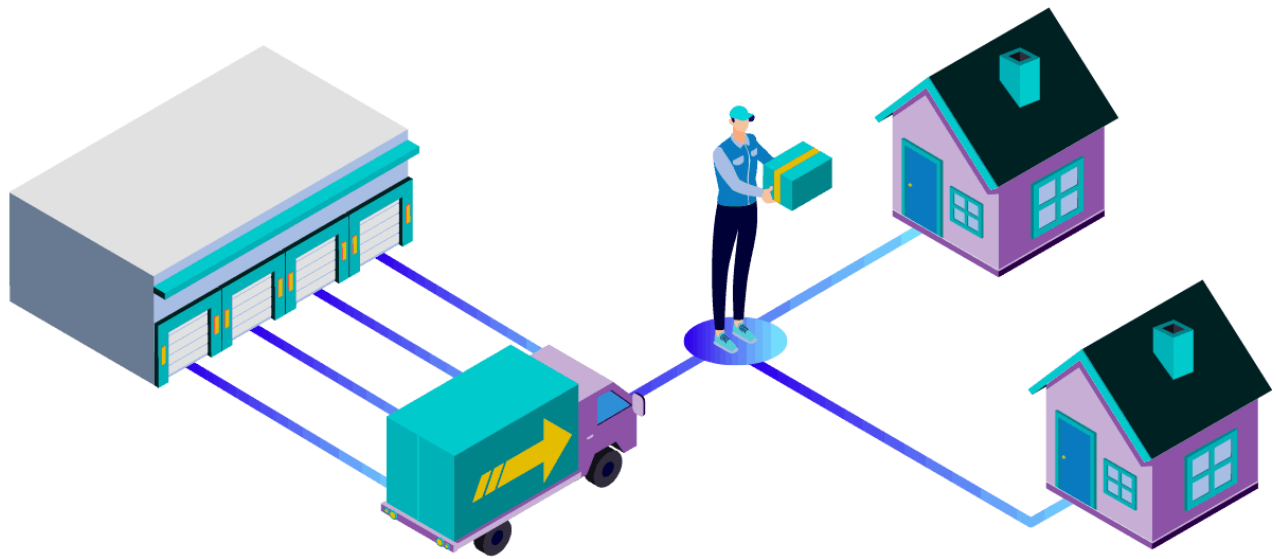
A **B2B model focuses on providing products from one business to another**. While many businesses in this niche are service providers, you'll find software companies, office furniture and supply companies, document hosting companies, and numerous other ecommerce business models under this heading.



B2B ecommerce examples you may be familiar with include the ExxonMobil Corporation and the Chevron Corporation, Boeing, and Archer Daniel Midlands. These businesses have custom, enterprise ecommerce platforms that work directly with other businesses in a closed environment. **A B2B ecommerce business typically requires more startup cash.**

B2c: Business To Consumer Ecommerce

The B2C sector is what most people think of when they imagine an ecommerce business. This is the deepest market, and many of the names you'll see here are known quantities offline, too. B2C sales are the traditional retail model, where a business sells to individuals, but business is conducted online as opposed to in a physical store.



Examples of B2C businesses are everywhere. Exclusively online retailers include Newegg.com, Overstock.com, Wish, and ModCloth, but other major B2C model brick-and-mortar businesses like Staples, Wal-Mart, Target, REI, and Gap.

C2C Ecommerce

B2B and B2C are fairly intuitive concepts for most of us, but the idea of C2C is different. What does a consumer-to-consumer ecommerce business look like?

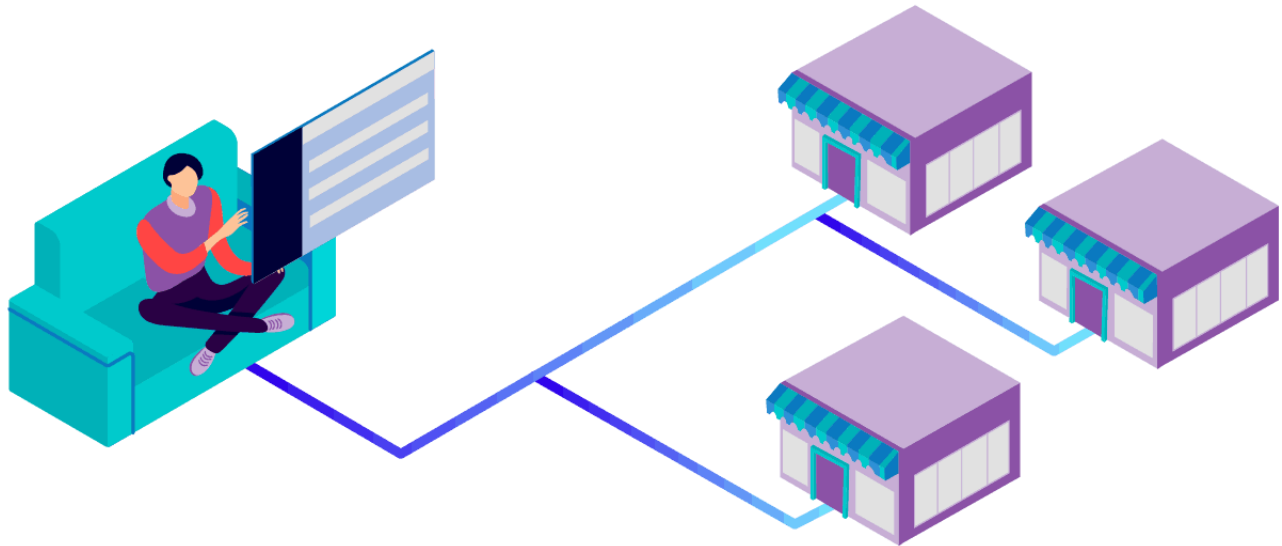


Created by the rise of the ecommerce sector and growing consumer confidence in online business, these sites allow customers to trade, buy, and sell items in exchange for a small commission paid to the site. Opening a C2C site takes careful planning.

Despite the obvious success of platforms like eBay and Craigslist, numerous other auction and classified sites (the main arenas for C2C) have opened and quickly closed due to **unsustainable models**.

C2B: Consumer To Business Ecommerce

C2B is another model most people don't immediately think of, but that is growing in prevalence. This online commerce business is when the consumer sells goods or services to businesses, and is roughly equivalent to a sole proprietorship serving a larger business.



Reverse auctions, service provision sites like UpWork, and **several common blog monetization strategies like affiliate marketing or Google AdSense** also fall under this heading.

Government / Public Administration Ecommerce

The models listed above are the primary ecommerce retail structures, but they aren't the only ones. Other types involve government/public administration conducting transactions with businesses or consumers.

- **B2G (also called B2A)**, for businesses whose sole clients are **governments or type of public administration**. One example is Synergetics Inc. in Ft. Collins, Colorado, which provides contractors and services for government agencies.
- **C2G (also called C2A)**: typically individuals paying the government for taxes or tuition to universities.

Two sectors that are closed for entrepreneur owners but are growing include G2B for government sales to private businesses, and G2C, for government sales to the general public.

1. Analysis:

The first step is understanding client requirement and the various dynamics around the clients existing systems, as the website or web application will eventually be integrated into this system. The analysis will then cover all the aspects of clients' business and their needs especially on how the website is going to be merged with their existing system.

The first important thing is finding the target audience. Then, all the present hardware, software, people, and data should be carefully assessed. For example, if a company XYZ Corp is in need of a website to have its Human Resource details online, the analysis team may look to review the existing data about the employees from the present database and what migration plan will be best suited to complete the transition.

The analysis should be done in the way, that it is neither too time to consume nor lacking information. The team should be able to come up with a detailed cost-benefit analysis. As the plan for the project will be an output of analysis, it should be realistic. To achieve, this the analyst should consult the designers, developers, and testers to come up with a realistic plan.

Input:

1. Kick off an interview with the client, initial Emails and supporting docs by the client, discussion notes,
2. Online chat transcripts, recorded telephone conversations, and
3. Model sites/applications

Output:

1. Work plan,
2. Estimating cost
3. Team requirements (No of developers, designers, QA, DBA etc)
4. Hardware-software requirements
5. Supporting documents and
6. Final client approval to go ahead with the project.

2. Specification Building:

Preliminary specifications are drawn up by covering up each and every element of the requirement. For example, if the product is a website then the modules of the site including general layout, site navigation and dynamic parts of the site should be included in the spec. Larger projects will require further levels of consultation to assess additional business and technical requirements. After reviewing and approving the preliminary document, a written proposal is prepared, outlining the scope of the project including responsibilities, timelines, and costs.

Input: Reports from the analysis team.

Output: Complete requirement specifications to the individuals and the customer/customer's representative (Technically, the project stakeholders)

Tools: For specification building, we recommend you to try [django-req](#) and a requirement management tool called [ReqView](#).

3. Design And Development:

After the specification building, work on the website is commenced after receipt of the signed proposal, a deposit, and any written content materials and graphics you wish to include. In this stage, the layouts and navigation will be designed as a prototype.

Some customers may be interested only in a fully functional prototype. In this case, we may need to show them the interactivity of the application or site. But in most of the cases, the customer may be interested in viewing two or three design alternatives with images and navigation.

Be prepared to note down quite a lot of suggestions and changes from the customer side. All the proposed changes should be finalized before moving into the next phase. The revisions could be redisplayed via the web for the customer to view.

This is the most vital stage in the project lifecycle to gain client trust that the project is in capable hands. Encourage customer comments, feedback and approvals to be communicated by e-mail, fax, and telephone. Engage in constant communication to give clients peace of mind.

Throughout the design phase, the team should develop test plans and procedures for quality assurance. It is necessary to obtain client approval on design and project plans.

In parallel, the Database team will assess and understand the requirements and develop the database with all the data structures. In this stage, the sample data will also be prepared.

Input: Requirement specification.

Output: Site design with templates, Images, and prototype.

Tools: There are plenty, The most popular one is Adobe Photoshop and SketchApp for Mac.

4. Content Writing:

This phase is necessary mainly for the websites. There are professional content developers who can write industry specific and relevant content for the site. Content writers need to add their text in such a way as to utilize the design templates. The grammatical and spelling check should be over in this phase. The type of content could be anything from simple text to videos.

Input: Designed template.

Output: Site with formatted content.

Tools: Grammarly

5. Coding:

Now, it's the programmer's turn to add his code without disturbing the design. Unlike traditional design, the developer must know the interface and the code should not disturb the look and feel of the site or application. This calls for the developer to understand the design and navigation of the site. If the site is dynamic then the code should utilize the template. The developer may need to interact with the designer, in order to understand the design. The designer may need to develop some graphics buttons whenever the developer is in need, especially while using some form buttons.

If a team of developers is working they should use a CVS to control their sources. The coding team should generate necessary testing plans as well as technical documentation. For example, Java users can use JavaDoc to develop their documents to understand their code flow. The end-user documentation can also be prepared by the coding team. This can be used by a technical writer who can understand them, and develop help and manuals later.

Input: The site with forms and the requirement specification.

Output: Database driven functions with the site, Coding documents.

Tools: For coding normally an IDE (Integrated Development Environment) will help you. Adobe Dreamweaver, PhpStorm, Netbeans etc are popular choices, and We can not really recommend the one – so feel free to choose yours.

6. Testing:

Unlike software development, web-based applications need intensive testing, as the applications will always function as a multi-user, multi-tier system with bandwidth limitations. Some of the testing which should be done are Integration testing, Stress testing, Scalability testing, load testing, resolution testing and cross-browser compatibility testing. Both automated testing and manual testing should be done without fail.

For example, it's vital to test fast loading graphics and to calculate their loading time, as they are very important for any website. There are certain testing tools as well as some Online testing tools which can help the testers to test their applications. For example, ASP.NET developers can use Microsoft's Web Application Test Tool to test the .NET applications, which is a free tool available from the Microsoft site to download.

After doing thorough testing at the live server too is necessary for websites and web-based applications. After uploading the site there should be a complete testing (E.g. Testing Links, Unit testing,)

Input: The site, Requirement specifications, supporting documents, technical specifications and technical documents.

Output: Completed application/site, testing reports, error logs, frequent interaction with the developers and designers.

Tools: There are aplenty. Just Google it.

7. SEO And Social Media Optimization:

This phase is applicable only for websites. Promotion needs preparation of meta tags, constant analysis and submitting the URL to the search engines and directories. The Search Engine Optimization and Social Media Marketing is normally an ongoing process as the strategies of the search engine may change quite often. Submitting a site URL once in 2 months can be an ideal submission policy. If the customer is willing, then paid click and paid submissions can also be done with additional cost.

Input: Site with unique and great content, Competitor study, keyword analysis.

Output: Site submission after necessary meta tag preparation.

Tools: <https://smallseotools.com/>, <https://www.seoptimer.com/fusionbd.com>

8. Maintenance And Updating:

Websites will need quite frequent updates to keep them fresh and relevant. In such a case, we need to do the analysis again, and all the other life-cycle steps will be repeated. Bug fixes can be done during the time of maintenance. Once your website is operational, ongoing promotion, technical maintenance, content management & updating, site visit activity reports, staff training and mentoring is needed on a regular basis, depending on the complexity of your website and the needs within your organization.

Input: The Web Site or Web Application, content/functions to be updated, re-Analysis reports.

Output: Updated application, supporting documents to other life cycle steps and teams.

Tools: For easy website maintenance CMS is the right choice. Investing in a CMS like WordPress or Joomla will make your site easy to maintain without much recurring expenditure.